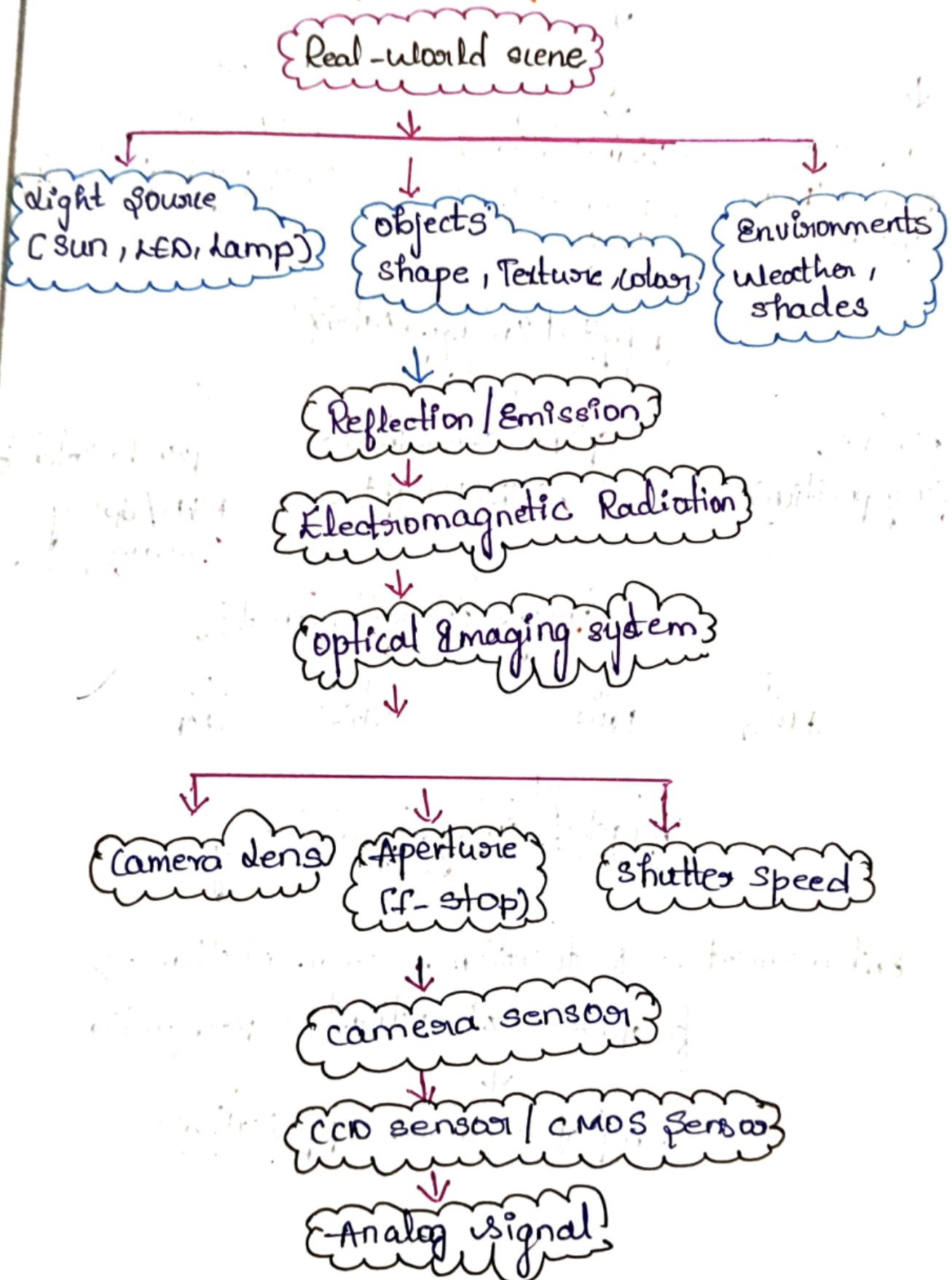
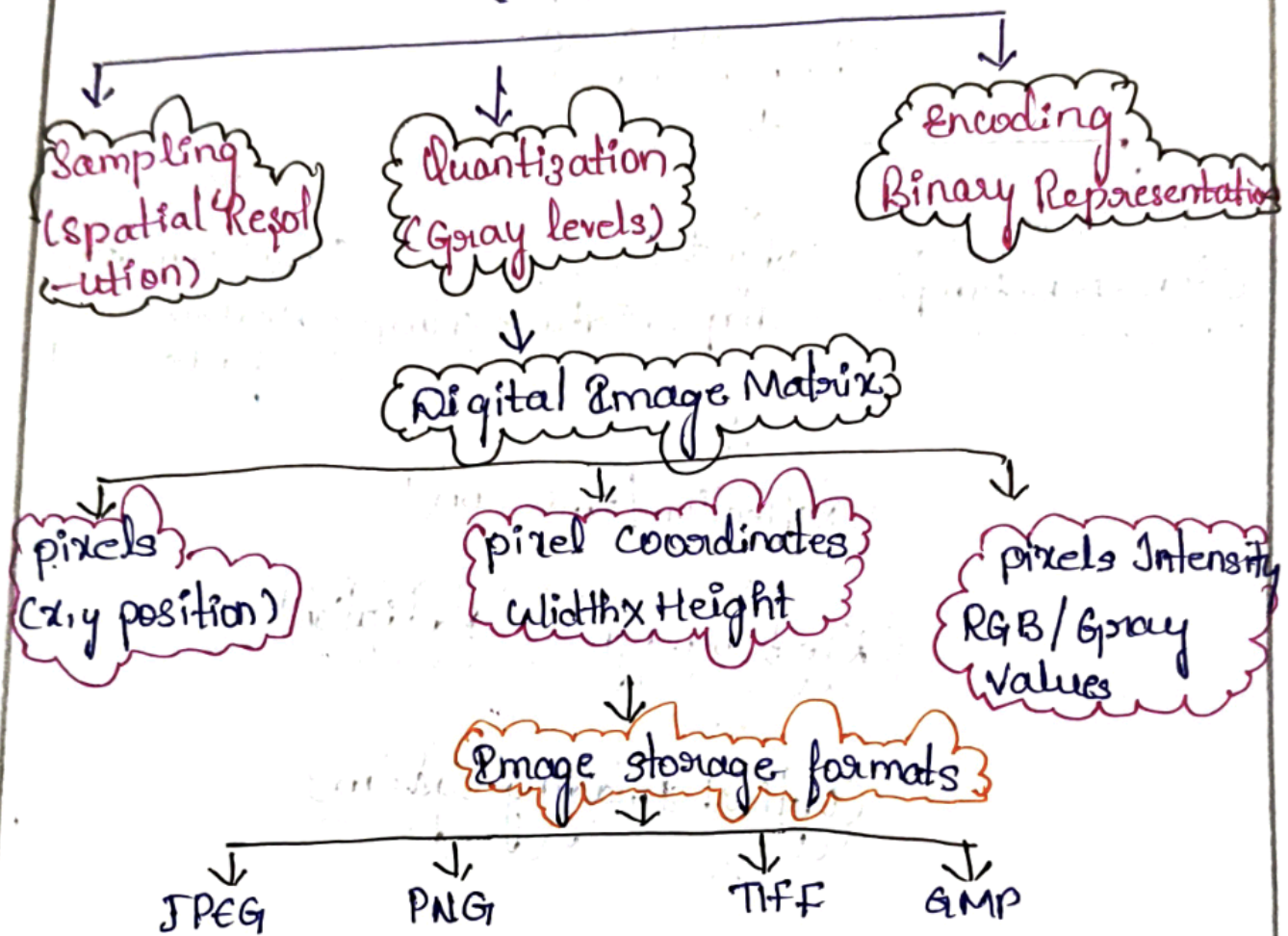


1. Comprehensive Concept Map: Real-World Scene Capture to Digital Image Representation



Analog to-Digital conversion

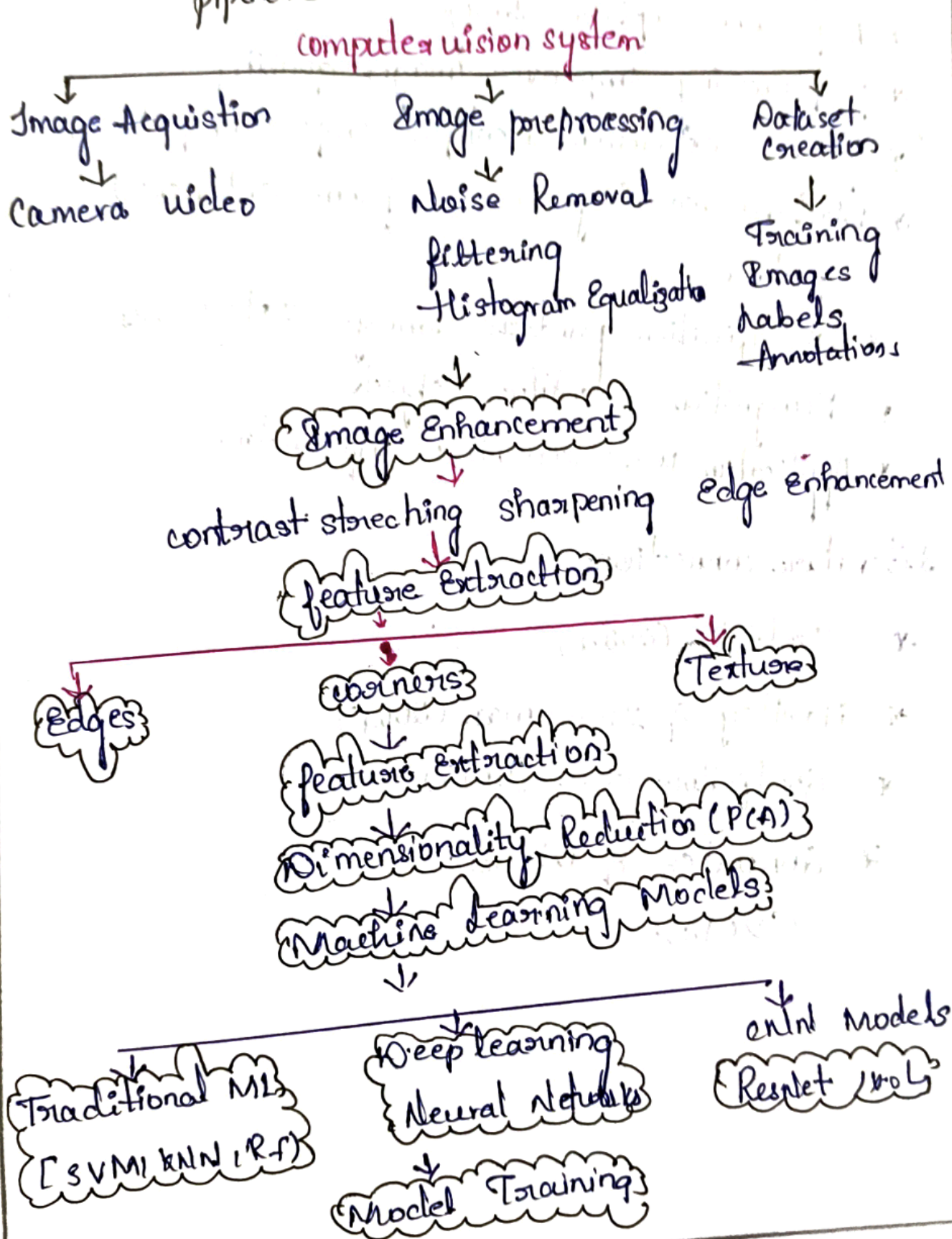


Digital Image processing

Enhancement → Restoration + Segmentation →
feature extraction → Recognition
↓
Computer vision Applications

2

2. Detailed Concept Map: computer vision system pipeline.



Relationships

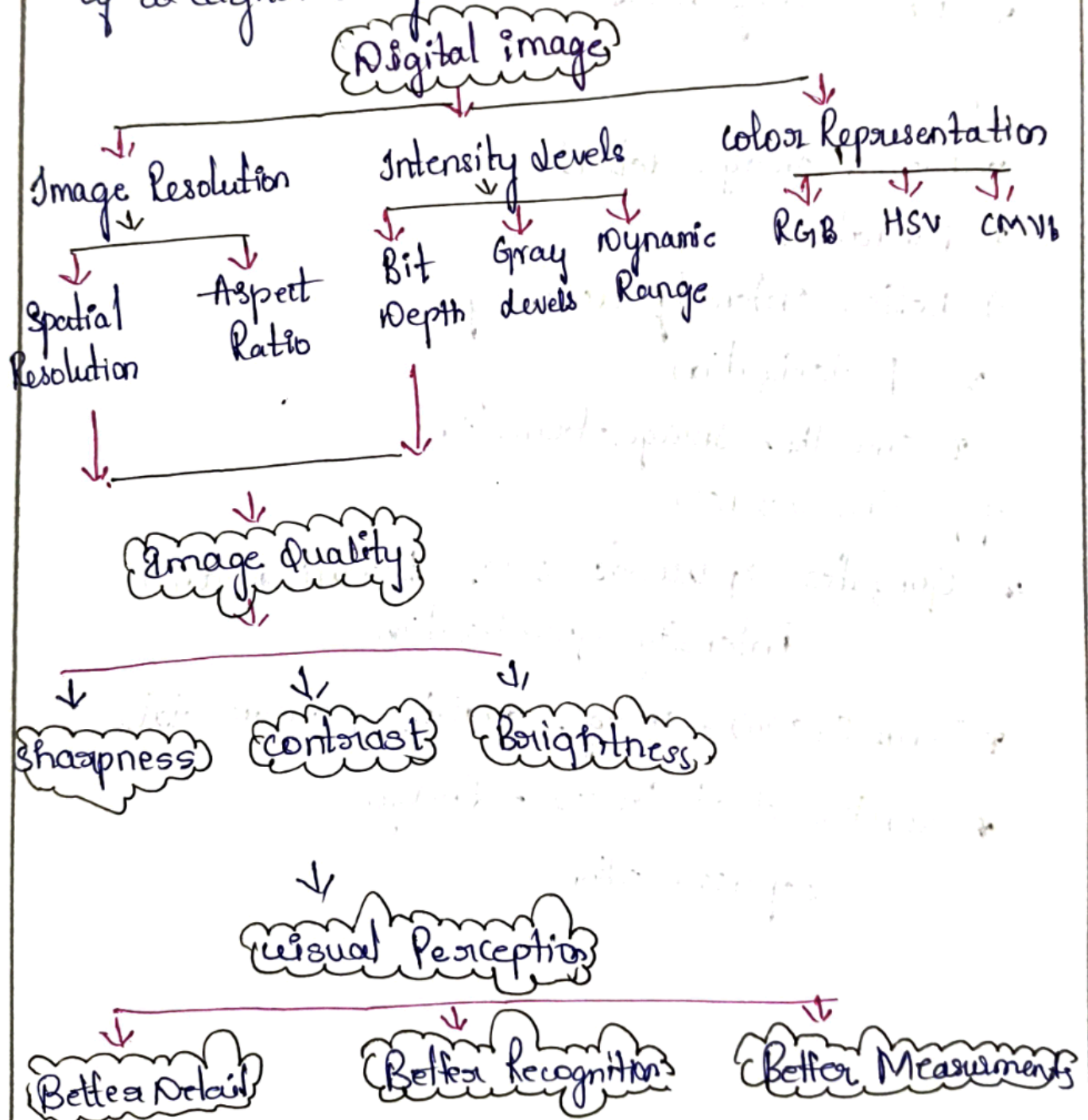
- * Acquisition capture images
- * preprocessing improves images quality
- * feature extraction identifies important patterns
- * Machine learning learns from extracted features
- * classification predicts image categories
- * Recognition identifies objects
- * Application use recognized information

literature connections

- * Szeliski (2022)
- * Hastley & Zisserman (2004)
- * Goodfellow et al (2016)
- * Gonzalez & Woods (2018).

3. Detailed concept map: Image Resolution and Intensity levels

* Image Resolution and intensity levels are key factors that determine the quality and clarity of a digital image.



* Cause and-effect Relationships

- * Higher Resolution
- * Higher Bit Depth
- * Low Bit Depth
- * More Gray levels

Effect

- * More image detail
- * Revelation
- * Better intensity accuracy
- * Posterization
- * Smoother image transitions

Supporting literature

- * Gonzales & Woods (2018) - Resolution and Intensity quantization
- * Jain (1989) - Image quality fundamentals
- * Sonka et al (2013) - Image Representation